

Time Series Analysis

Introduction

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Time Series Analysis

Introduction

Types of statistical data

- **Cross sectional data**

- Many objects at one time (example: election polls)

- **Time series data**

- One objects observed in many consecutive units of time (years, quarters, months, days, minutes, seconds, ticks etc.) (example: asset listed on a stock exchange)
 - Time series is a chronologically ordered set of value of the analysed feature or specific phenomenon observed in different moments (intervals) of time

- **Panel data**

- Many objects observed in many consecutive units of time (example: GDP changes for countries from the EU)

Time series data examples

- Time series data examples

- **Economics**

- industrial production (y, q, m), private consumption (y,q), government expenditures (y), CPI (y,q,m), PPI (y,q,m), money supply (m,w)

- **Finance**

- exchange rate (m,d,hf), interest rate (m,d,hf), asset price (m,d,hf), stock exchange index (m,d,hf), sentiment changes (m,w,d) etc.

- **Other**

- weather conditions (d,hf), popularity rankings (m,w,d) etc.

Modelling approaches

- **Time series models**

- Modelling current value of a variable based on its own history.
- TSM account for the fact that data points taken over time may have an internal structure (such as autocorrelation, trend and/or seasonality) that should be accounted for.
- Examples: (S)ARIMA, GARCH models.

- **Econometric models**

- Modelling current value of a depended variable based on independent variables value (current & past)
- Examples: DL, OLS models.

- **Hybrid**

- Best of the two worlds
- Examples: ARDL, ARIMAX, ECM.

Applications of TSM

- **Steps in modelling**

- Understanding an underlying process that produced observed data
- Apply appropriate method to fit a model
- Forecast future values of a process better than by simple guessing

- **Examples of modeling TS (plain and hybrid)**

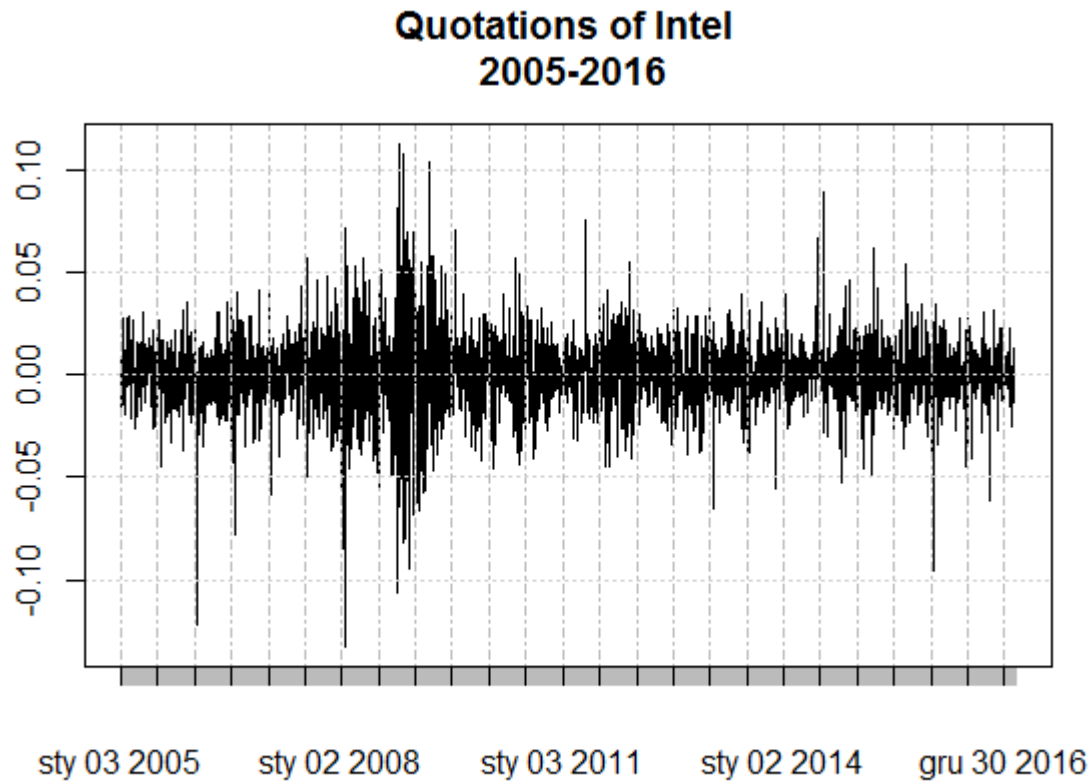
- Building automated trade strategies
- Risk modeling
- Market efficiency analysis
- Price forecasting
- Demand forecasting
- Economy growth
- Etc.



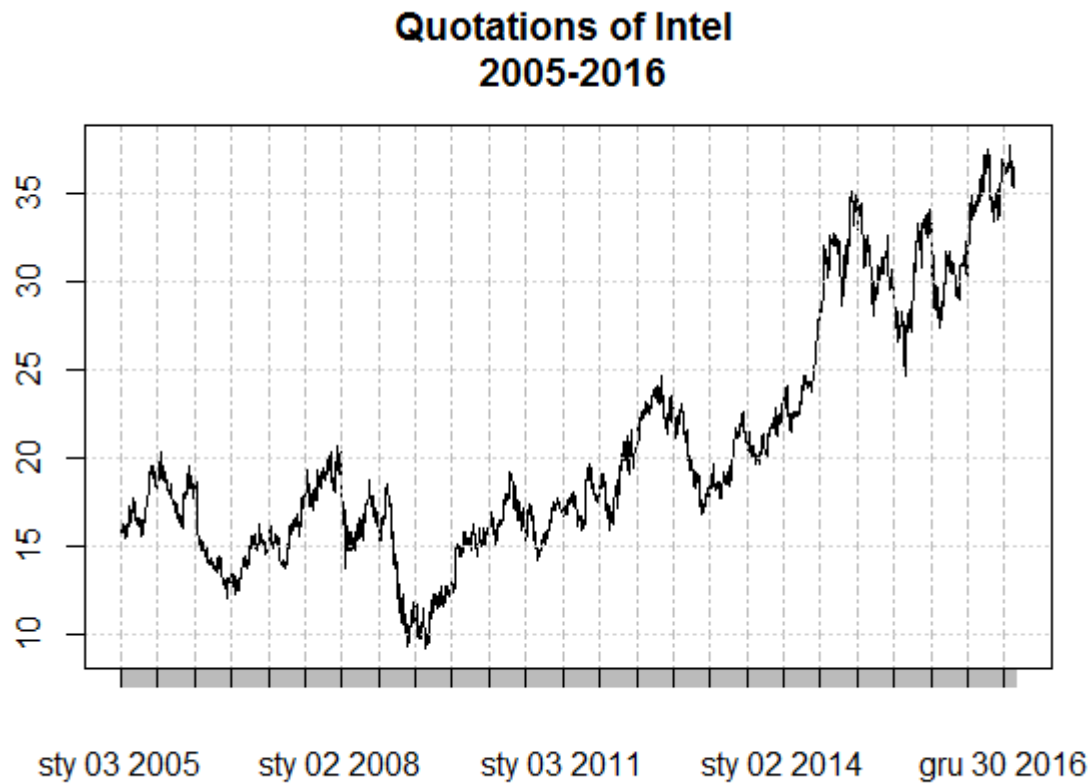
Stationarity

- Key concept of TSA
- Roughly speaking
 - a time series is stationary if its behaviour is not changing over time
 - its value always vary around the same level and variability is constant over time
- Stationary series have a rich theory -- their behaviour is well understood
- Not all time series that we analyse are stationary
- Non-stationary series tend to be the rule rather than the exception.

Stationary process



Non-stationary process



Time Series Analysis

Introduction to forecasting



Introduction to forecasting

- Forecast
 - an attempt to anticipate the future development of the analyzed phenomenon (direction and/or amount).
 - main goal of TSA.
- Forecasting is useful!
 - Sometimes it is enough to predict the direction of change to earn money, eg. tomorrow's quotations for stocks.
 - On the other hand model predicting that prices will increase (CPI) is rather useless -- for last 40 years decrease happened in 1% of cases.

Famous bad forecasts

- **Everything that can be invented has been invented.** Charles H. Duell, Commissioner, U.S. Office of Patents, 1899
- **Stocks have reached what looks like a permanently high plateau.** prof. Irving Fisher, Yale University, 16/10/1929
- **I think there is a world market for maybe five computers.** Chairman of IBM, 1943
- **Computers in the future may weight no more than 1.5 tons.** Popular Mechanics, forecasting the relentless march of science, 1949
- 640K ought to be enough for anybody. Bill Gates, 1981
- Drill for oil? You mean drill into the ground to try and find oil? You're crazy! Drillers who Edwin L. Drake tried to enlist to his project to drill for oil in 1859
- **Who the hell wants to hear actors talk?** H.M. Warner, Warner Brothers, 1927
- I have travelled the length and breadth of this country and talked with the best people, and I can assure you that data processing is a fad that won't last out the year. The editor in charge of business books for Prentice Hall, 1957

Concept of forecast accuracy

- None forecast is perfect, each involves an error!
 - Even if all model assumptions are met and all statistical procedures applied correctly, there appears a **random error** of the forecast
 - Expected forecast error can be calculated (*ex-ante* error)
 - In huge majority of cases a real forecast error is higher than expected.
 - Fallacy: a model which does not meet the assumptions can be useful in forecasting.
- When forecast is good (better)?
 - Forecasts might be useful if they provide better answers than alternative methods or simple guessing
 - Quality test of the forecast should not be based on its faultlessness, but on comparison of its accuracy with other forecasts

Bad and inaccurate forecasts

- Reasons for bad forecasts
 - **structural change** in model parameters
 - wrong model **specification**
 - **missing, smoothed or incorrect data**
 - change in **expectations** of market players
 - change in **policy**
 - **accumulation** of errors in **long-term** forecast
- Reasons for big random error in forecasts
 - Variables do not have a normal distribution
 - Residuals are not independent
 - Values of independent variables (factors) are not known at the time of forecast calculation
 - Data generating process changes over time

Alternative types of forecasts

- Alternative types of forecasts
 - point or interval
 - unconditional or conditional
 - forecast scenarios with probabilities
 - in-sample and out-of-sample
 - one-period or multi-period
 - short-term or long-term
 - for one variable or several (many) variable



How to assess forecast accuracy?

- *Ex-ante*
 - describes how well the forecast **might be** (difficult to calculate)
- *Ex-post*
 - describes how well the forecast really **was**
- *Ex-post* forecast error as a proxy for *ex-ante*
 - From the whole data period select a subsample of observations **in-sample** period.
 - Estimate the model on that subsample.
 - Generate the forecast for the **out-of-sample** data.
 - Calculate forecast error for the **out-of-sample** period.

$$MAE \text{ (Mean Absolute Error)} = \frac{1}{T - T_s} \sum_{t=T_s+1}^T |y_t - y_t^*|$$

$$MAE \text{ (Mean Square Error)} = \frac{1}{T - T_s} \sum_{t=T_s+1}^T (y_t - y_t^*)^2$$

$$MAPE \text{ (Mean Absolute Percentage Error)} = \frac{100\%}{T - T_s} \sum_{t=T_s+1}^T \left| \frac{y_t - y_t^*}{y_t} \right|$$

$$AMAPE \text{ (Adjusted MAPE)} = \frac{100\%}{T - T_s} \sum_{t=T_s+1}^T \left| \frac{y_t - y_t^*}{y_t + y_t^*} \right|$$

TSA vs Structural models in forecasting

- Structural modelling started in the 1920s & 1930s.
 - Early attempts to build forecasting models centred on the estimation of equations to predict agricultural prices and quantities and stock market prices.
 - The discipline then spread to building models for macroeconomic forecasting purposes.
- While early models included only 20 to 30 equations, their size rapidly increased, and by the 1970s some macro models contained thousands of individual equations.
- Problems with sophisticated structural models
 - While initial expectations for these macroeconomic models were quite optimistic, the quality of forecasts they delivered was disappointing
 - The worst performance occurred in the 1970s and early 1980s, when models were unable to predict the simultaneous occurrence of double-digit inflation and unemployment.
 - In particular, these models were also unable to predict any of the four recessions occurring between 1970 and 1982.
- Sources of error in structural models
 - These failures, coupled with the inability of econometric models to predict financial markets accurately, led to:
 - a re-examination of the sources of error in structural models,
 - a movement away from this approach toward **non-structural models**.
 - Error from structural models comes from:
 - random elements,
 - frequent shifts of underlying data generation function, which make it difficult to predict the independent variables accurately.



Non-structural models

- for these reasons, econometric analysis and development of forecasting models has shifted in the past 30 years toward placing a greater emphasis on non-structural models.
- In these non-structural models, a particular variable is a function only of:
 - its own lagged variables,
 - lagged error terms,
 - truly exogenous variables such as time trends.
- Hybrids (TSA + some explanatory variables)
- Motivation of time series models
 - It is quite likely that some combination of methods (structural models, time series models, and exogenous information) → can provide more accurate forecasts than any single method.
 - In most cases, time-series models perform better when the trend has been removed from the original data.
 - Hence, we first turn to a discussion of those issues before focusing on the estimation of autoregressive (AR) and moving average (MA) models.

Time Series Analysis

Objects in R

Selected date and time index classes in R

CLASS	PACKAGE	DESCRIPTION
Date	base	Represent calendar dates as the number of days since 1970-01-01
POSIXct, POSIXlt	base	Represent calendar dates as the (signed) number of seconds since the beginning of 1970 as a numeric vector. Supports various time zone specifications (e.g. GMT, PST, EST etc.)

ts objects

- *ts* object
 - Regularly spaced time series data, that are **separated by a fixed interval** of time, may be represented as objects of class **ts**,
 - Such data are typically observed monthly, quarterly or annually,
 - **ts** objects are created using the `ts()` constructor function (base R).
- important limitation of *ts* objects
 - The **ts** class is rather limited, especially for representing financial data that is not regularly spaced.
 - For example, the **ts** class **cannot be used to represent daily financial data because such data are only observed on business days**.
 - That is, a business day time clock generally runs from Monday to Friday skipping the weekends.
 - So data are equally spaced in time within the week but the spacing between Friday and Monday is different.
 - This type of irregular spacing cannot be represented using the **ts** class.

zoo objects

- A very flexible time series class is *zoo* (Zeileis' ordered observations) created by Achim Zeileis and Gabor Grothendieck and available in the package *zoo* on CRAN.
 - **zoo** class was designed to handle time series data with an arbitrary ordered time index.
 - this index could be a regularly spaced sequence of dates, an irregularly spaced sequence of dates, or a numeric index.
 - a **zoo** object essentially attaches date information with data.
- *zoo* object creation
 - to create a **zoo** object one needs a **time index and data**.
 - the time index must have the same number of rows as the data object and can be any vector containing ordered observations.
 - typically, the time index is an object of class Date, POSIXct, POSIXlt or observation number.

xts objects

- *xts* class is the abbreviation from the *extensible time-series* class
 - at the core of an ***xts*** object is a zoo object.
 - what motivated the ***xts*** extension was a desire to have even more flexibility, while imposing reasonable constraints to make this class into a true time-based one.
 - the most important difference between zoo and ***xts*** is that the index of each ***xts*** object **must be of a known and supported time or date class** (Date, POSIXct, POSIXlt, chron).